# Replication of Study 2 by Markowitz (2024, PNAS Nexus)

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2026-11-10

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### Introduction

I am interested in understanding the psychological implications of generative AI, including how people perceive and respond to AI-generated information. I am particularly interested in whether using AI to make information more accessible changes not only people's understanding of that information but also their perceptions of its source, the author. Scientific information is one area where information may be difficult to grasp for non-experts because the language can be complex. I therefore chose to replicate Study 2 of Markowitz (2024), "From

Complexity to Clarity: How AI Enhances Perceptions of Scientists and the Public's Understanding of Science" and to extend it with an additional condition. In Study 2, Markowitz experimentally tested whether reading simpler AI-generated scientific summaries rather than more complex human-written summaries of the same papers changed perceptions of scientists' trustworthiness, credibility, and intelligence. In Study 3, Markowitz expanded this study with more stimulus pairs and two comprehension measures (a multiple choice question to capture the gist of each summary and a free response summary). This work aligns with my broader interest in how generative AI may change people's psychological responses to information and to those who communicate it.

Study 1 consisted of gathering the PNAS significance statements, generating AI versions of the statements using GPT-4, and compared their linguistic features, including common words, analytic writing, and readability. In Study 2, the researcher selected five pairs of human-written significance statements and GPT-generated significance statements with the largest difference in common words. Participants read summaries from three of the five pairs, and for each selected pair they were randomly assigned to the GPT (simple) or PNAS (complex) version. Stated another way, each participant had the possibility of seeing three entirely AI-generated summaries or entirely human-written summaries from the way the original study was designed. After reading each summary, participants completed manipulation checks assessing its clarity, complexity, and how much of the summary they understood. Participants would then assess the 'scientist's' trustworthiness, credibility, and intelligence and indicate whether they believed the summary was written by a human or by AI. Participants were not told which summaries were AI-generated. A future line of work the author states is to study how people judge scientific summaries that are labeled as AI-generated compared to non-disclosures.

Study 2 found small effects that went in different directions, where GPT summaries were rated as more credible and more trustworthy, but their authors were rated as less intelligent and the three-item perceptions index (trustworthiness, credibility, and intelligence) was only modestly higher for GPT summaries. Following feedback from the PSYC 201A teaching team, my single confirmatory test is the effect of summary version (GPT vs. PNAS) on ratings of the scientist's *trustworthiness*, using the original linear mixed model. I chose trustworthiness for a few reasons. First, it was the largest author-perception effect in Study 2. Second, it is the outcome most closely tied to the paper's motivating claim that simpler AI writing can counter declining trust in scientists. Third, it did not replicate in Study 3, which used 20 stimulus pairs, and the author suggests that the multitude and difference in the stimuli of Study 3 may explain this difference. My replication uses the same five pairs as Study 2, where I ask a narrower question of does the Study 2 trust effect replicate with the same stimuli in a new sample? I am not using the three-item perceptions index as the confirmatory outcome because it averages items that have effects in different directions (GPT 'authors' were rated more trustworthy and credible but less intelligent).

In the original study, participants did not know that some summaries were AI generated which suggests that the Study 2 effect is an effect of simpler writing, not of knowing that AI was involved. There is an interesting and emerging line of research on judgments of AI users

and AI-generated content and I am interested in whether knowing AI was used to generate text would change people's perceptions of the source. I will therefore add a between-subjects *disclosure* factor with two arms. Participants are randomly assigned to one arm, and summary version stays randomized within each participant, as in the original. The *no-label arm* will run exactly like the original Study 2. No summary has a label here. The confirmatory test uses only this arm. The *label arm* is where every AI summary will include the label "AI-generated summary of the author's abstract" directly below the text explanation. The human written summaries do not include any labels which mirrors how real disclosure policies work where only AI generated content is specified (though the human written summaries receiving no label is a potential confound. I will discuss this in my limitations section).

This extension to the original study aims to investigate whether disclosure changes the GPT vs PNAS difference in trustworthiness. As the teaching team suggested, I plan to run a simulation-based power analysis before committing to the disclosure arm. If I cannot recruit enough participants to power both the replication and the extension, then I will prioritize the replication and I may drop the disclosure arm or run as a pilot study with a small number of participants.

To address additional comments from the teaching team, I aim to randomly assign the AI-generated or human written statements in balancing these versions within participants so that no participant recieves either all human-written statements or all AI-generated statements. As for concerns about the time frame for completing this study, I plan to add two more summaries (five per person, the same number as Study 3) and Study 3's multiple-choice and free-response comprehension measures for each summary. I am also adding a short AI-use questionnaire. My estimate is about 15-20 minutes, which I will confirm with the timing of the pilot study.

To analyze this data, I will use linear mixed-effects models because each participant provides multiple observations and each stimulus is seen by many participants. Futher following Markowitz (2024), the confirmatory model includes summary version (GPT/simple vs PNAS/complex) as the fixed effect and random intercepts for both participant and stimulus. Because there are only five stimuli, the stimulus random intercept is estimated from very few levels and may be estimated at or near zero (a singular fit). The analysis plan below specifies in advance how I will handle that. Also to note, with five stimuli, any conclusion applies to these five summary pairs rather than to scientific summaries in general.

One anticipated challenge will be to gather enough participant data in the time frame of a few weeks, depending on the sample size from the power analysis. Another challenge is the difference in samples. The original study drew from a sample of U.S. adults on Prolific. For this replication project, SONA participants consist of undergraduate students, who may be very familiar with AI tools and AI-generated outputs. Because of this familiarization with AI, SONA participants' assessments may lead to variations in their ratings and human/AI judgments such that they may use AI tools much more and may be better at spotting AI-stylistic outputs. To measure this, I aim to add questions about how often participants use AI chatbots and how confident they are at detecting AI-written text, which I will use in exploratory analyses. Another challenge is that generative AI tools have increasingly improved

in their capabilities from 2024 to 2026. This means that the 2024 outputs might be more recognizable due to the nature of AI tool outputs at the time. Now, AI generated outputs may differ, where people may find it more difficult to distinguish AI-generated text from human-written text, because current models produce writing that is more varied and natural-sounding than the 2024 outputs. To stick to the original study, I plan to use Markowitz's original GPT-generated summaries as stimuli rather than generating new ones.

For my learning goals, I want to learn how to concretely run a power analysis. I am also interested in simulating the results of this study, especially since I need to do so anyways to determine the effect sizes since the original paper did not include them. I hope to develop a deeper understanding of coding in R. Regarding feasibility, I think if I stick with the original AI-generated summary statements, it will be feasible to replicate this experiment, except with more summary statements and adding an AI-disclosure condition.

Link to repo: https://github.com/sisegall/markowitz2024

Linked pathway to the original paper: https://github.com/sisegall/markowitz2024/blob/main/original_paper/Markowitz_2024_PNASNexus_AI_science_summaries.pdf

Original data and materials (OSF): https://osf.io/64am3/ · Study 2 data: https://osf.io/download/xjnfg/ · Stimuli: https://osf.io/download/fnh74/ · Preregistration (Study 2): https://aspredicted.org/C3K_T31

## Methods

### Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

***For the analysis, I will first rerun the original trustworthiness model on the OSF data to check that I can reproduce the effect of* t* = 4.63, and it gives me the fixed-effect and variance estimates for the `simr` power analysis that will set the sample size.***

### Planned Sample

The power analysis will determine the sample size per arm (no-label and AI-label). I will stop collecting data when the target number of complete participants is reached in each arm, or at the end of the class deadline for the data-collection window, whichever comes first. If the class deadline for the data collection window closes first, I will report the achieved sample size and the power it gives under the simulation model.

Participants will be UCSD undergraduates recruited through SONA who are 18 or older and they will receive course credit. The participants should be native English speakers. I will collect age, gender, race/ethnicity, and political ideology, as the original Markowitz (2024) study did, as well as exploratory questions related to AI-use and confidence in AI detection. I expect the sample to be younger and use AI tools more than the original Prolific sample. Participants will be assigned to the no-label or label arm with condition assignment, which will keep the arms balanced as data come in.

## Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

## Procedure

**Stimuli.** The stimuli are the five GPT/PNAS significance-statement pairs from Study 2, which the original paper describes as "Five pairs of stimuli from study 1b were selected for the experiment, having had the greatest difference in common words scores between the PNAS and GPT texts." I will use the texts verbatim from the original OSF stimulus file (osf.io/64am3). The papers are PNAS 1507704112 (adenosine/dopamine receptors), 1809962115 (subduction zone temperatures), 1902227116 (heparan sulfate in sepsis), 2004900117 (titin oxidation in heart muscle), and 2220518120 (DNA in a bacteriophage capsid).

Every measure from Study 2 is used with the original wording and 7-point scales: - *Manipulation checks:* "How clear was the writing in the summary you just read?" and "How complex was the writing in the summary you just read?" (1 = not at all, 7 = extremely); "How much of this writing did you understand?" (1 = not at all, 7 = an enormous amount). - *Author perceptions:* "How intelligent is the scientist who wrote this summary?", "How credible is the scientist who wrote this summary?", and **"How trustworthy is the scientist who wrote this summary?"** (the confirmatory outcome) (1 = not at all, 7 = extremely). - *Author identity:* "This summary was written by a human." and "This summary was written by Artificial Intelligence." (1 = strongly disagree, 7 = strongly agree). - *Demographics:* age, gender, race/ethnicity, and political ideology (1 = extremely liberal, 7 = extremely conservative).

**Added from Study 3. (exploratory, end of study additions)** After all five summaries and their Study 2 ratings are done, participants complete a comprehension section at the very end of this study. Participants first read and rate all five summaries using only the Study 2 measures. Only then do they answer the additional add-ons from Study 3. So, for each of the five summaries, in random order, they see the paper's topic as a cue (e.g., "The summary about [heparan sulfate in sepsis]"), then the multiple-choice question form the Study 3 OSF stimulus file (answer options presented in random order), then the Study 3 free-response prompt: "In

your own words, please summarize the main idea and findings of the study you just read about. Be as detailed as possible." The summary text is not shown again.

**Added for the extension and the sample difference (new).** - *Disclosure label (label arm only).* A label appears directly above the text, inside the summary box, on every GPT summary and only on GPT summaries. The wording is: "AI-generated summary of the author's abstract." This label appears on every screen that shows that GPT-summary. PNAS summaries will not have a label. The no-label arm has no labels on any summary. - *Disclosure manipulation check (both arms, after all summaries):* "Did any of the summaries you read have a label saying it was generated by AI?" (Yes / No / Not sure). - *AI use (both arms):* "How often do you use AI chatbots such as ChatGPT, Claude, or Gemini?" (Never, Less than once a month, About once a month, About once a week, Several times a week, Daily) and "How confident are you that you can tell whether a piece of writing was produced by AI?" (1 = not at all, 7 = extremely). - *Attention check (both arms, in the final questionnaire):* "To show that you are reading carefully, please select 'Strongly disagree' for this question."

Every study follows this order: read summary, comprehension checks, rate summary on perceptions index (even though I only care about the trustworthiness), record their specific AI use, AI detection confidence, [other exploratory items from AI survey if need be], disclosure manipulation check (if that is the condition participants are assigned to), then demographics.

### Analysis Plan

**Clarify key analysis of interest here** You can also pre-specify additional analyses you plan to do.

(Before data collection I still need to refit the original trustworthiness model to check that I reproduce the reported result ($t = 4.63$), using the original OSF data (titled `experimental_data_OSF.csv`). Note to self that I will need to refit it to only the 210 participants who saw both versions, since this is closer to my balanced design, and use the fitted parameters in the power analysis.)

**Confirmatory model**:

$$\text{Trust}_{ij} = \beta_0 + \beta_1 \,\text{Version}_{ij} + u_{0i} + w_{0j} + \varepsilon_{ij}$$

i = indexes participants j = indexes the five stimulus pairs Version = +0.5 is GPT, -0.5 is PNAS _1 = GPT - PNAS for difference in mean trust

lmer(trust ~ version_c + (1 | paid) + (1 | stimulus))

**Singular fit.** With only five stimuli, the stimulus variance may be estimated at or near zero, which lme4 reports as a singular fit. If that happens, I will refit the model with stimulus as a fixed effect (four dummy codes) instead of a random intercept, keep the participant random intercept, and use that model for the confirmatory test. I will report both models. Either way, my concusions apply to these five summary pairs, not to scientific summaries in general.

**Extension model.**

$$\text{Trust}_{ij} = \beta_0 + \beta_1 \,\text{Version}_{ij} + \beta_2 \,\text{Arm}_i + \beta_3 \,\text{Version} * \text{Arm}_{ij} + u_{0i} + w_{0j} + \varepsilon_{ij}$$

**Data inclusion and exclusion rules.** The original preregistration states, "No outliers nor exclusions are anticipated," and the paper reports no exclusions. To match this, the confirmatory analysis includes all participants who completed the study, which matches the original (no exclusions). As a robustness check, I will rerun the same confirmatory model on a restricted sample that exlcudes participants who (a) failed the attention check, or (b) spent under 10 seconds on the reading screen for 3 or more of the 5 summaries. If the two analyses disagree, I will report both and treat the confirmatory analysis with no exclusions as the primary analysis.

**Confirmatory analysis (single key test).** The outcome is trustworthiness ratings (1-7), in the **no-label arm only**, which is the direct replication. The predictor is summary version. I will test $\beta_1$ two-tailed at $= .05$, using degrees of freedom from `lmerTest` as in the original analysis (the original cites lme4 and lmerTest). I will report $\widehat{\beta}_1$, its SE, df, $t$, $p$, and 95% CI, along with the estimated marginal means for each version, the variance components, and marginal and conditional $R^2$ (`MuMIn`, as in the original).

**Replication criterion.** The study replicates if $\widehat{\beta}_1 > 0$ and $p < .05$, meaning GPT summaries are rated as more trustworthy than PNAS summaries. I will also report whether the original estimate (about 0.28) falls inside the 95% CI of the replication estimate.

**Secondary analysis: disclosure extension** (to run ONLY if the disclosure arm is kept after the power analysis..). The key term is the interaction, which tests whether labeling GPT summaries as AI-generated changes the difference in trust. I predict that disclosure could lower trust in labeled texts, which is consistent with aversion to AI-generated content overall.

### Differences from Original Study

**Sample:** The replication will run with UCSD undergraduates on SONA instead of U.S. adults on Prolific. Students likely use AI tools more and are perhaps more prone to recognize AI writing, which could weaken the GPT advantage, change the human/AI judgments, or both. The paper does not identify sample characteristics as moderators. AI use is measured so this can be examined.

**Setting:** Online in both cases. SONA participants receive course credit instead of payment.

**Number of summaries and assignment:** Participants see five summaries instead of three, and versions are balanced within participants instead of assigned independently for each pair. Every participant contributes both versions, which should increase precision. The extra trials could add fatigue, but the measures are short and Study 3 also used five summaries.

**Added comprehension measures:** Study 3's multiple choice and free response questions are added in a separate section after all summaries and ratings, and the consent form doesn't announce a comprehension test. This keeps the Study 2 procedure unchanged for the confirmatory outcome. The cost is that comprehension is measured from memory rather than right after each summary so scores may be lower and noisier and are not directly comparable to Study 3's findings (that had the MCQ and FRQs right after each summary). Summary order and version are randomized so memory loss should affect AI and human summaries equally and should not bias the comparison between them.

**AI-disclosure arm:** This is new and between-subjects, and the confirmatory test uses only the no-label arm, which matches the original.

**Added end-of-study measures:** the disclosure check, AI-use items, and the attention check come after all ratings, so they cannot affect the confirmatory outcome.

**Analysis:** The confirmatory model matches the original model. The original analyzed several outcomes, while for this project trustworthiness is the single confirmatory outcome.

**Time since original:** Participants in 2026 may judge 2024 GPT-4 text differently from participants in 2024. I am keeping the original stimuli so that I can run a direct replication of Study 2.

### Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

### Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

### Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

## Results

### Data preparation

Data preparation following the analysis plan.

### Confirmatory analysis

The analyses as specified in the analysis plan.

*Side-by-side graph with original graph is ideal here*

### Exploratory analyses

Any follow-up analyses desired (not required).

## Discussion

### Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

### Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.